Sharp dimension bounds for right-angled hyperbolic polytopes

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Abstract

We prove that a finite-volume right-angled hyperbolic polytope has dimension at most eight, and at most four if all its vertices are ideal. Both bounds are attained. The proofs share an elementary projection: from a finite interior point, project to a facet of maximum unit-normal height. Repeating this operation three times in dimension nine gives a six-face whose wall directions form a spherical code with pairwise products strictly below $1/4$. A cubic tensor-square identity bounds this code by 26, contradicting the published minimum of 27 facets. For the ideal bound, one projection in dimension five combines Musin's kissing-number theorem and Kolpakov's 24-cell rigidity; an opposite-normal calculation excludes the metric equality case.

1 Statements and geometric inputs

A polytope is *entirely ideal* if all its vertices are ideal. Throughout, hyperbolic polytopes have finitely many facets and finite volume. A right-angled polytope has dihedral angle $\pi/2$ at every actual ridge. Finite and ideal vertices are otherwise allowed in any combination, and no arithmetic hypothesis is imposed.

Theorem 1. *A finite-volume right-angled hyperbolic polytope has dimension at most eight.*

Theorem 2. *An entirely ideal finite-volume right-angled hyperbolic polytope has dimension at most four.*

Finite-volume right-angled examples in dimension eight are recalled in [5]. The regular ideal right-angled 24-cell gives the four-dimensional example. Thus both bounds are sharp. The exclusion proofs below use three published numerical statements, isolated here with their precise scopes.

Proposition 3 (Six-dimensional facet minimum)**.** *A six-dimensional finite-volume right-angled hyperbolic polytope has at least* 27 *facets.*

This is the consequence of Lemma 1 and Proposition 4 of Dufour [2] explicitly recalled in [1, Section 4]. It applies to every vertex type, rather than only entirely ideal polytopes. We need neither a classification nor an equality statement for 27 facets.

Proposition 4 (Musin)**.** *A finite subset of $S^3 \subset \mathbb{R}^4$ with pairwise products at most $1/2$ between distinct members has cardinality at most* 24*.*

This is the Main theorem of [4, p. 4]. We use its cardinality bound, not uniqueness of a kissing configuration.

Proposition 5 (Kolpakov)**.** *An entirely ideal right-angled hyperbolic four-polytope has at least* 24 *facets. Equality holds only for the regular ideal right-angled* 24*-cell, up to hyperbolic isometry.*

This is Theorem 4 of the published paper [3, p. 1945]. The last two propositions are needed only for Theorem 2. The spherical bound used for Theorem 1 is proved directly in Section 3.

We use standard geometric foundations for acute-angled polytopes. In the Lorentz model, distinct outward unit facet normals have nonpositive products, including facets that do not share a ridge. As explained in [6, p. 42, §2.1], actual nonobtuse ridge angles imply the all-wall condition: nonadjacent supporting planes do not intersect, and the off-diagonal Gram entries are nonpositive. Faces of dimension at least three inherit finite volume and right angles. Entire ideality is also inherited by faces. No projection chain below reaches a two-face, where ideal polygon angles would require separate treatment.

2 Descending maximum facet height

Use Lorentz signature $(-,+,\dots,+)$, future unit timelike points, and outward *unit* spacelike normals. Thus

$$P = \{x \in \mathbb{H}^n : \langle x, m\rangle \le 0 \text{ for every actual facet normal } m\}, \qquad \langle m, m'\rangle \le 0 \quad (m \ne m').$$

All heights below refer to this fixed unit-normal scaling. They are the hyperbolic sines of distances to the corresponding walls.

Lemma 6 (Farthest-facet projection)**.** *Let q be a finite relative-interior point of a right-angled polytope P, with $q^2 = -1$. Set $t(m) = -\langle q, m\rangle > 0$, choose a facet normal n with largest height $t = t(n)$, and put*

$$q' = \frac{q + tn}{\sqrt{1+t^2}}. \tag{1}$$

Then q' lies in the relative interior of the actual facet $Q = P \cap n^\perp$. Every intrinsic facet normal m_i of Q is an actual ridge neighbor of n, remains unit, and has height

$$0 < t_i' = -\langle q', m_i\rangle = \frac{t(m_i)}{\sqrt{1+t^2}} \le \frac{t}{\sqrt{1+t^2}} < 1. \tag{2}$$

Proof. The numerator in (1) has square $-1-t^2$ and product zero with n. Its product with q is $-1-t^2 < 0$, so normalization gives a future unit point in $n^\perp$. For *any* other actual facet normal m,

$$\langle q', m\rangle = \frac{\langle q, m\rangle + t\langle n, m\rangle}{\sqrt{1+t^2}} < 0. \tag{3}$$

The first numerator term is strict and the second is nonpositive. Thus q' lies in the actual facet and is strict against all other walls; it is not merely a formal point on the supporting plane.

An intrinsic facet of Q comes from its unique ridge neighbor m_i. Right angles give $\langle n, m_i\rangle = 0$, so restriction to $n^\perp$ preserves its unit norm. Formula (2) now follows from the maximum choice of t. □

If all heights at q are at most A, monotonicity of $x/\sqrt{1+x^2}$ gives a new bound $A/\sqrt{1+A^2}$. Start with the finite maximum A_0 at any finite interior point. Repeating the construction intrinsically gives

$$A_j \le \frac{A_0}{\sqrt{1+jA_0^2}} < \frac{1}{\sqrt{j}} \qquad (j \ge 1), \tag{4}$$

whenever the selected faces still have dimension at least three. The formula follows by induction, or by adding one to the inverse square of the previous bound. Strictness holds even if the maximizing facet is not unique at a step.

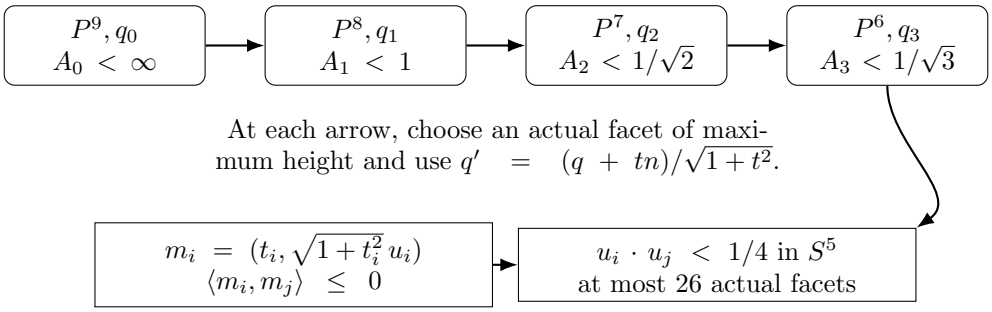


Figure 1: Three actual facet choices and the resulting spherical code. The arrows are the exact projection formulas, not ordinary edges of the polytope. The final code has one direction for each actual facet; the tensor calculation below contradicts the published minimum of 27.

We will use the following translation from heights to spherical codes. At a unit point q of a d-dimensional face, choose q as its time axis and write the intrinsic unit normals as

$$m_i = \big(t_i, \sqrt{1+t_i^2}\, u_i\big), \qquad u_i \in S^{d-1} \subset \mathbb{R}^d. \tag{5}$$

For distinct facets, all-wall acuteness gives

$$u_i \cdot u_j \le \frac{t_i t_j}{\sqrt{(1+t_i^2)(1+t_j^2)}}. \tag{6}$$

The directions are distinct: if $u_i = u_j$, the Lorentz product of the two unit normals would be positive. Consequently the number of code points is exactly the number of facets of the selected face.

3 A strict spherical bound in six variables

Lemma 7. *A finite set of unit vectors in $\mathbb{R}^6$ whose distinct-pair products are strictly less than $1/4$ has at most 26 members.*

Proof. Consider the cubic polynomial

$$F(x) = (x + \tfrac{1}{2})^2 (x - \tfrac{1}{4}) = x^3 + \tfrac{3}{4}x^2 - \tfrac{1}{16}. \tag{7}$$

We give a direct tensor-square proof of the required positive-kernel identity, so no spherical linear-programming theorem is needed.

Use ordinary Euclidean tensor inner products. For a unit vector u define the rank-two tensor

$$A(u) = u \otimes u - I/6$$

and the symmetric rank-three tensors

$$S(u)_{abc} = u_a \delta_{bc} + u_b \delta_{ac} + u_c \delta_{ab}, \qquad T(u) = u^{\otimes 3} - S(u)/8.$$

For $x = u \cdot v$, contraction gives

$$\langle A(u), A(v) \rangle = x^2 - \tfrac{1}{6}, \qquad \langle T(u), T(v) \rangle = x^3 - \tfrac{3}{8}x. \tag{8}$$

Each cross contraction of a cubic tensor with S is $3x$; the remaining contraction is

$$\langle S(u), S(v) \rangle = (3 \cdot 6 + 6)x = 24x.$$

The scalar identity therefore becomes

$$F(u \cdot v) = \tfrac{1}{16} + \tfrac{3}{8}(u \cdot v) + \tfrac{3}{4}\langle A(u), A(v)\rangle + \langle T(u), T(v)\rangle.$$

Thus, for $u_1, \dots, u_N$,

$$\begin{aligned}\sum_{i,j} F(u_i \cdot u_j) &= \frac{N^2}{16} + \frac{3}{8}\|\sum_i u_i\|^2 \\ &\quad + \frac{3}{4}\|\sum_i A(u_i)\|^2 + \|\sum_i T(u_i)\|^2 \\ &\geq \frac{N^2}{16}. \end{aligned} \tag{9}$$

On the other hand, $F(1) = 27/16$, and $F(x) \leq 0$ for $x < 1/4$. The diagonal terms and (9) give

$$\frac{N^2}{16} \leq \sum_{i,j} F(u_i \cdot u_j) \leq \frac{27N}{16},$$

so $N \leq 27$. If $N = 27$, equality throughout forces each off-diagonal polynomial value to vanish. Since $u_i \cdot u_j < 1/4$ strictly, the only available root is $-1/2$. Four such vectors would already give

$$\|u_1 + u_2 + u_3 + u_4\|^2 = 4 + 12(-\tfrac{1}{2}) = -2,$$

which is impossible. Hence $N \leq 26$. □

Proof of Theorem 1. Suppose first that a right-angled nine-polytope exists. Three projections from a finite interior point produce an actual six-face R and a relative-interior point at which every facet height is strictly less than $1/\sqrt{3}$, by (4). The inherited face is right-angled and has finite volume. For its intrinsic normals, (6) now gives

$$u_i \cdot u_j < \frac{1}{4} \qquad (i \neq j)$$

in $S^5 \subset \mathbb{R}^6$. Lemma 7 gives at most 26 facets for R, contradicting Proposition 3. Every higher-dimensional polytope has a nine-face inheriting finite volume and right angles, giving the same contradiction. □

4 The ideal bound from one projection

We first record the small metric calculation needed at equality. In the standard Lorentz model of the regular ideal right-angled 24-cell, its outward unit normals are

$$(1, \epsilon e_i + \delta e_j), \qquad 1 \leq i < j \leq 4, \quad \epsilon, \delta \in \{1, -1\}, \tag{10}$$

and the center is $(1, 0)$.

Lemma 8. *No future unit timelike point of the regular ideal right-angled 24-cell has all its unit-normal heights strictly less than one. More precisely, if every height is at most one, the point is the center and all the heights equal one.*

Proof. The normals in (10) come in opposite spatial pairs $(1, r), (1, -r)$. Write the point as (a, w), with $a > 0$ and $a^2 - \|w\|^2 = 1$. The two corresponding heights are $a - r \cdot w$ and $a + r \cdot w$. If both are at most one, their sum gives $a \leq 1$. Future unit timelikeness gives $a \geq 1$. Thus $a = 1, w = 0$, and every height equals one. Strict inequalities on all heights are impossible. □

Proof of Theorem 2. Assume an entirely ideal right-angled five-polytope exists. A single farthest-facet projection from a finite interior point gives a four-facet Q at whose relative-interior point all heights satisfy $0 < t_i < 1$. It inherits finite volume, entire ideality and right angles. By (6), its spherical directions in S^3 have pairwise products strictly below $1/2$. Proposition 4 bounds its facet number by 24; Proposition 5 gives the reverse bound and identifies Q with the regular ideal 24-cell. Its metric equality case, not a uniqueness assertion for kissing configurations, is what is used here. Lemma 8 contradicts the strict heights $t_i < 1$.

There is no entirely ideal right-angled five-polytope. In any higher dimension a five-face would inherit all three hypotheses and give the same contradiction. The ideal right-angled 24-cell attains dimension four. □

Both proofs use actual facets and a finite point checked against every wall. Neither infers a polytope from a set of formal Gram vectors, nor infers finite volume from a few feasible vertices. The strict inequalities come from projecting a finite point; they require no limiting or generic-position argument. For the general bound the only numerical geometric input is the six-dimensional facet minimum. For the ideal bound the additional inputs are exactly Musin's cardinality theorem and Kolpakov's metric equality theorem. All remaining calculations are displayed above.

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